

WINMOL stem segmentation — method corrections and scale-augmentation results

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1 Executive summary

Two independent bodies of work, in the order they should be read.

1.1 Part I — Corrections to the published WINMOL method

Read directly from WINMOL_segmentor (R) and WINMOL_Analyzer (Python), with every value quoted from source or measured by running it. Five items change how the published results should be read, and three of them are places where the papers and the code disagree.

#	Finding	Consequence
1	The composite F1/BCE loss does not train. F1Score_loss calls the <i>metric</i> , which applies k_round; a rounded value has zero derivative almost everywhere.	The model trains on plain BCE . The term appears in the reported loss and is absent from the optimisation. Verified three ways, including an 8-epoch R run agreeing with pure BCE to eight decimal places .
2	The split is a random 80/20 over tiles , drawn per stage — not site-level, not spatial. With 100× oversampling, tiles overlap.	Early stopping, the LR schedule and checkpoint selection all read an optimistic validation signal.
3	Tile side length is 15 m, not ≈15.4 m , fixing effective GSD at 2.93 cm/px .	Load-bearing: matching training GSD to it was worth +14.6 F1 in our measurements.
4	Node spacing is 50 cm, not the documented 25 cm (min(min_length, 0.5)).	Measured 0.47–0.50 m on real output across four orthomosaics.
5	A transposed digit in the decoder (265 filters where the symmetric ladder gives 256).	The published .hdf5 is 31,140,642 parameters; the described architecture gives 31,036,673. Not reproducible from the paper without the typo.

Plus two coordinate traps in the vectoriser that produce plausible geometry rather than an error — the skeletonisation pad is never removed, and the EDT diameter branch reads pixel (row, col) as easting/northing. Full detail with file and line references in Part II.

A reproduction worth noting. Trained single-stage on the published SpecDS and scored on the published TestDS with our own metric code: **0.7581**, which matches the 2022 paper’s *best two-stage pretrained* result (75.6%) without any GenDS pretraining. The published beech model scores **0.6812** on that same TestDS — below the paper’s figure, unexplained, and worth raising with the author.

1.2 Part II — Scale augmentation

The Analyzer’s effective resolution is $\text{tile_size} / 512$, a user-set knob most people never touch deliberately. Jittering the training footprint $\pm 30\%$ makes a model robust to it:

	fixed spread	jitter spread	improvement	cost at 1.00×
HRNet, within-site	0.0295	0.0207	−0.0087 (t −6.0)	−0.0011 (noise)
UNet, within-site	0.0334	0.0146	−0.0188 (t −57.0)	+0.0020 (3/3 seeds)
UNet, clean site holdout	0.1231	0.1116	−0.0115	+0.0328 (3/3 seeds)

Recommendation: enable it by default. It costs nothing at the deployment scale, removes 30–56% of the degradation when the tile size moves, and on the one clean site holdout where segmentation works at all it is worth **+3.3 F1** at the deployment scale.

It does **not** replace matching the training scale to the serving scale — that is worth 14.6 F1 against this 0.9–3.3. Do both.

1.3 Part III — Bachsee_north is not a bad site

Every model trained here scores $F1 < 0.13$ on Bachsee_north, and the corpus notes have long attributed this to its amber November canopy. Three explanations were tested; the first two are wrong.

checked	result
Are the labels misregistered? Annotations are EPSG:25833, ortho EPSG:32633 — a datum pair diverging ~0.5 m.	No. On the orthomosaic’s own grid every beech site peaks at zero offset; residual under 4 cm.
Are the stems invisible under closed canopy?	No. That came from a <i>luminance-only</i> statistic. In CIELAB, Bachsee’s stem/background separation is 1.75 — second highest in the corpus, above Campus’s 1.16 , and Campus scores F1 0.53.
Is the site learnable at all?	Yes. Trained on 252 tiles from its own west half and tested on the spatially separated east half: F1 0.6203 .

252 of its own tiles beat 3,588 tiles from every other site by 53 F1 points. The data is sound; the failure is **domain shift**.

Re-running the folds with Bachsee **excluded from training** (18 runs) confirms it belongs there: overall **-0.045 F1**, 5 of 18 paired comparisons positive. The damage is ordered by phenological distance from Bachsee’s November amber — Campus (October) **-0.095**, Campus_Oberheide (leaf-off February) **-0.045**, Kaufland (high summer) **+0.005** — and the two Campus folds lose the identical 800 tiles, so tile count does not explain the ordering. The out-of-domain site is the corpus’s phenological diversity, not noise. Bachsee is the corpus’s only autumn-phenology site, so when it is held out nothing in training has shown the model a stem against orange foliage. **Strong hue augmentation fixes it.** Randomising hue over the full circle (—aug—hue—shift

90 --aug-sat-shift 60 --aug-hsv-p 0.9) takes the Bachsee holdout from **F1 0.042 to 0.688** (3/3 seeds) for **-1.7 F1** in domain (Kaufland, 3/3 seeds). The base arm predicts almost nothing there — precision 0.87–1.00, recall 0.00–0.03; the strong arm reaches recall 0.58–0.66. This retires the standing advice that a site holdout is impossible on this corpus. It is an **interaction, not a dose**: a full hue circle at the default probability reaches only 0.261, and half the hue range at full probability only 0.177, while both together reach 0.688. Do not soften the setting.

An earlier claim in this repo that fold F1 is monotone in stem contrast came from the luminance-only statistic and is **withdrawn**.

1.4 What is not settled

- **The site holdout rests on one informative fold.** Of four beech sites, two are the same forest 4.4 years apart (33% footprint overlap) and one is Bachsee_north, which is outside every training domain (below).
- **Label scarcity.** Under 1 ha of stem is labelled in total; beech is near exhausted and pine has none.

2 Data inventory

Every number here was measured from the files on 2026-08-14, not transcribed from earlier notes. Rasters read with rasterio, vectors with fiona; areas are in each layer's own CRS units (m²). Source tree: /Volumes/storage/Datasets/Winmol/training_data/WINMOL_Trainings_Data/GIS (mounted on T14 as /data/mnt/storage/...).

2.1 The headline number

0.98 ha of labelled stem exists in total, across 8 annotated sites. 29 orthomosaics are held; **21 of them have no stem annotations at all**. Labelling, not imagery, is the binding constraint on this project.

2.2 1. Orthomosaics — all 29

Year is the acquisition date encoded in the filename (YYYYMMDD).

2.2.1 Beech (10 rasters)

acquisition	site	GSD	size	CRS
2017-10-16	EW_WW_Campus	6.39 cm	147 MPx	EPSG:32633
2017-11-14	EW_WW_Bachsee_north	4.08 cm	214 MPx	EPSG:32633
2017-11-23	EW_WW_Bachsee_south	3.95 cm	257 MPx	EPSG:32633
2021-07-06	EW_WW_Kaufland	2.09 cm	125 MPx	EPSG:25833
2022-02-26	EW_WW_Campus_Oberheide	3.36 cm	1911 MPx	EPSG:25833
2022-02-26	EW_WW_Campus_Oberheide (clip)	3.36 cm	373 MPx	EPSG:25833
2023-11-06/07	Altmann_Jauerling (AT)	3.30 cm	540 MPx	EPSG:3416
2024-07-19	FR17203_Abt- 128_129_30	3.52 cm	280 MPx	none
2025-03-06	172_03_Kohlleiten_Meislingeraamt (AT)	3.52 cm	280 MPx	EPSG:3416
2025-03-27	171_4_Kraking_Windwurf	3.99 cm	76 MPx	EPSG:25833

2.2.2 Spruce (13 rasters)

acquisition	site	GSD	size	CRS
2022-02-09	Bremerhagen 1–8	1.23 – 1.84 cm	302 – 802 MPx	EPSG:25833
2022-02-12	Barnekow 1, 3, 5, 6	1.51 – 1.76 cm	297 – 870 MPx	EPSG:25833
2022-02-12	Barnekow 2, 4 (IR)	1.75 / 2.07 cm	70 / 141 MPx	EPSG:25833
2022-03-09	EW_WW_Kaltes_Wasser_2	1.64 cm	815 MPx	EPSG:25833
2022-03-09	EW_WW_Ruheforst	1.61 cm	796 MPx	EPSG:25833

2.2.3 Pine (3 rasters)

acquisition	site	GSD	size	CRS
2022-03-09	EW_WW_Stromtrasse	2.75 cm	625 MPx	EPSG:25833
2022-03-29	Alt_Madlitz_plot123	2.58 cm	603 MPx	EPSG:25833
2022-03-29	Alt_Madlitz_plot4	3.24 cm	134 MPx	EPSG:25833

GSD spans 1.23 – 6.39 cm/px, a 5.2× range. The Analyzer serves at a fixed `tile_size / 512` (2.93 cm/px by default), so most of this corpus is nowhere near the serving scale unless resampled — which is what `--extent/--tile-px` extraction does.

2.3 2. Stem annotations — only 8 sites carry them

acquisition	site	species	polygons	stem area	AOI	CRS
2017-10-16	Campus	beech	822	2,908 m ²	118,808 m ²	32633
2017-11-14	Bachsee_north	beech	136	451 m ²	31,527 m ²	25833
2021-07-06	Kaufland	mixed	125	250 m ²	5,393 m ²	25833
2022-02-26	Campus_Oberniedel	mixed	490	1,917 m ²	111,631 m ²	25833
2022-02-09	Bremerhagen_spruce	spruce	158	180 m ²	2,566 m ²	25833
2022-02-12	Barnekow_3	spruce/pine	358	741 m ²	15,516 m ²	25833
2022-02-12	Barnekow_5	spruce/pine	1,012	1,826 m ²	27,788 m ²	25833
2022-02-12	Barnekow_6	spruce	741	1,528 m ²	<i>none</i>	25833
total			3,842	9,802 m² = 0.98 ha		

A ninth file, `202171114_EW_WW_Bachsee_north`, is a **typo'd duplicate** of the 2017-11-14 layer — the same 136 polygons and 451 m². It has no matching orthomosaic. Ignore it; do not “fix” the name and pair it, or those stems are counted twice.

2.3.1 Species composition of the labelled polygons

code	species	polygons
GFI	Gemeine Fichte (Norway spruce)	1,986
RBU	Rotbuche (European beech)	1,211
GKI	Gemeine Kiefer (Scots pine)	231
DGL	Douglasie (Douglas fir)	97
GBI	Gemeine Birke (birch)	10
RGU	Roterle/other	1
—	no species attribute	~295

No pine site is annotated. The 231 GKI polygons are pine stems inside the two Barnekow spruce stands, not a pine acquisition — all three pine orthomosaics have zero labels.

“Beech” sites are not pure beech. Campus_Oberheide is GFI 199 / RBU 190 / DGL 92, so the beech models in this repo are trained on a mixed stand that happens to sit in the beech folder. Kaufland is RBU 110 / GBI 10 / DGL 5.

2.4 3. The four beech sites used for training

These are the sites behind every result in `scale-augmentation-results.md` and `scale-augmentation-los0.md`.

site	date	season	GSD	hue	saturation	notes
Campus	2017-10-16	autumn	6.39 cm	15°	0.19	coarsest raster in the corpus
Bachsee_north	2017-11-14	late autumn	4.08 cm	25°	0.73	amber canopy; CRS mismatch vs its ortho
Campus_Oberheide	2020-02-26	leaf-off	3.36 cm	32°	0.25	33% footprint overlap with Campus
Kaufland	2021-07-06	high summer	2.09 cm	98°	0.32	finest raster; smallest AOI

Two facts that shaped the experiments:

- **Campus and Campus_Oberheide are the same forest**, imaged 4.4 years apart; 33% of Campus’s tiles overlap a Campus_Oberheide footprint (centroids 183 m apart). They are not independent sites, so a “leave-one-site-out” fold on either is really a temporal holdout.
- **Bachsee_north is the only late-autumn acquisition**, at 2.4–3.8× the saturation of every other site. Held out it scores F1 ~0.09; trained on its own west half it reaches **0.62** on its east half. Removing it from training costs the other folds up to 9.5 F1.

2.5 4. Published datasets (Zenodo record 5682580, Reder et al.)

Verified byte-identical against the record.

dataset	tiles	tile size	on disk	role
GenDS10	4,540	2000×2000	2.3 GB	synthetic stage-1 pretraining
SpecDS	454	313×313	12 MB	species-specific stage-2
TestDS	117	313×313	3.2 MB	the papers’ test set

Two caveats measured here. **GenDS10’s 2000 px tiles carry roughly 500–650 px of real information** — a round trip through 512 costs 1.45 grey levels against SpecDS’s 3.73, so they are largely interpolation; effective GSD ≈ 4.6 cm/px. And **TestDS contains 117 tiles, not the 106 stems the 2022 paper describes**.

GenDS10_512 (4,540 tiles, 736 MB) is our pre-resized copy: the loader’s exact resize applied once instead of every epoch, verified to reproduce the loader’s array to 0.0000/255.

2.6 5. Datasets built in this repo

All carry `tiles.jsonl` (source ortho, world centre, rotation, GSD, stem fraction), which is what makes any tile traceable back to the ground via `scripts/locate_tile.py`.

dataset	tiles	tile px	extent	GSD	purpose
BeechScale6663,388 / 400 / 400		666	19.512 m	2.9297 cm	scale-augmentation arms
loso/site_all/Campus	800	666	19.512 m	2.9297 cm	LOSO fold source
loso/site_all/Campus_Oberhaid	800	666	"	"	"
loso/site_all/Bachsee_north	800	666	"	"	"
loso/site_all/Kaufland	388	666	"	"	"
loso/site_tv/{Campus,Oberhaid} each		666	"	"	block train/val

Kaufland yields only 388 tiles because its AOI is 5,393 m² and a 19.512 m footprint holds 13.80 m in from every edge.

2.7 6. Defects to handle

- **Mixed CRS.** Campus and Bachsee north/south orthos are EPSG:32633; everything else is EPSG:25833 — the same UTM zone on different datums, so a silent mismatch shifts masks by metres rather than failing. 20171114_EW_WW_Bachsee_north has annotations in 25833 and its ortho in 32633. Both scripts reproject explicitly, and this was **verified**: every beech site's polygons peak at zero offset on its own raster grid (`scripts/check_registration.py`), residual under 4 cm.
- **One orthomosaic has no CRS at all** (20240719_FR17203_Abt-128_129_30.tif). It is unannotated, so nothing depends on it yet.
- **202171114_EW_WW_Bachsee_north** — typo'd duplicate, see above.
- **Species case variants:** RBU/rBU/RBu, GFI/GFi/gFI, DGL/DGL/dGL, plus single-record typos GIF and GFIU. Match case-insensitively or silently drop records. ~295 polygons carry no species at all.
- **No DEM/DSM/CHM anywhere** in the corpus — height is not available as a channel.
- **Two Barnekow rasters are infrared** (2 and 4), not RGB; they are unannotated.
- **gends10_ready** is a locally derived copy that is not trusted; use the Zenodo originals.

3 Reder's method: gaps closed from source, and where the docs disagree with the code

The methodological report on Reder et al. 2022 / 2025 lists items "NOT recoverable from openly accessible sources" and points at WINMOL_segmentor (R) and WINMOL_Analyzer (Python) as the authoritative places to read them. Both were read directly. This closes what is closeable and flags three places where the published description and the code disagree.

Every value below is quoted from source or measured by running it, with the file and line.

3.1 1. Hyperparameters — closed

item	value	source
optimizer	Adam, lr = $1e-3$, β_1 0.9, β_2 0.999, decay = 0, amsgrad = FALSE	controlling.R
batch size	4	main_training.R
epochs	100 cap per stage	training.R
early stopping	val_loss, patience 3 (stage 1) / 5 (stage 2)	callbacks.R:15–17
LR schedule	ReduceLROnPlateau, factor 0.1, patience 2, min_delta = $1e-4$	callbacks.R:7–12
checkpoint	save_best_only = TRUE on val_loss	callbacks.R:6
augmentation	flips + brightness/contrast/saturation/hue, train split only	input_pipeline.R

3.2 2. Split construction — closed, and it is not site-level

```
train_data <- initial_split(train_data, prop = 0.8) # training.R, both stages
```

A **random 80/20 split over tiles**, drawn independently per stage. Not spatial, not site-level. The report infers a site-aware evaluation from the 15-site design; the training code does not implement one. With the generator’s 100× oversampling, tiles overlap, so this validation signal is optimistic — which matters because early stopping, the LR schedule and checkpoint selection all read it.

3.3 3. The composite loss does not train — measured

Both papers present the F1/BCE composite as the key choice for foreground/background imbalance. In code:

```
F1Score <- custom_metric("F1Score", function(y_true, y_pred) {
  y_pred <- k_round(y_pred) # <- hard threshold
  ... })
F1Score_loss <- function(y_true, y_pred) {
  loss_binary_crossentropy(y_true, y_pred) + (1 - F1Score(y_true, y_pred)) }
```

F1Score_loss calls the **metric**, and the metric rounds. k_round has zero derivative almost everywhere, so the F1 term contributes no gradient: **the model trains on plain binary cross-entropy**. The term is real in the reported loss value and absent from the optimisation.

Verified three independent ways:

- **In torch** — R’s expression transcribed gives a gradient torch.equal to plain BCE’s (grad-norm 0.012018 for both), while the reported value is ~2× larger.
- **In R itself** — LOSS=f1 (Stefan’s untouched controlling.R) against loss_binary_crossentropy alone, same seed, 8 epochs: precision, recall and F1 agree to **eight decimal places** at every epoch. Only loss differs, by exactly $1 - F1$.
- **Ablated** — replacing it with a *differentiable* soft-F1 term, paired over seeds, shifts recall **+3.8** and precision **−2.5** (4/4 seeds, paired t 4.86 / −5.47). So the term does something once it has a gradient; in the published form it does nothing.

This does not invalidate the results — BCE is a reasonable loss — but the stated rationale for the loss choice is not what the model experienced.

3.4 4. Architecture — a transposed digit

model_UNet.R:87,91, decoder stage E7:

```
layer_conv_2d(filters = 265, ...) # every other stage is symmetric: 512, 256, 128, 64
```

The transpose conv above it correctly emits 256. Both convs of that block use **265**. The published .hdf5 loads at **31,140,642** parameters; the symmetric ladder gives 31,036,673. Harmless numerically, but “31M U-Net” is not reproducible from the paper’s description without the typo.

3.5 5. Instance-extraction defaults — closed

classes/Config.py:

parameter	default	GUI label in the report
min_length	2.0 m	Min Length
max_distance	8	Max Distance
tolerance_angle	7	Maximum Angle
max_tree_height	32 m	Maximum Tree Height
tile_size	15 m	Tile side length
img_width	512 px	Tile pixel extent
overlap_pred	8	—
stem_binary_threshold	0.5	—

Tile side length is 15 m, not ≈ 15.4 m. The report derives 15.36 m from 3 cm $\times$ 512 px; the code fixes 15 m, giving an effective GSD of **15/512 = 2.93 cm/px**. That number is load-bearing: matching training GSD to it was worth **+14.6 F1** in our own measurements.

Skeletonisation (utils/Skeletonization.py): `skimage.morphology.skeletonize` on the binary mask, after padding by `int(max_tree_height / px_size) + 1`. Endpoints and branchpoints are found by crossing-number (transitions == 1 endpoint, ≥ 3 branchpoint); branchpoints are removed, dense nodes eroded, then segments refined to the measuring-point spacing with a minimum length of `floor((min_length/4) / px_size)`.

Diameter: two methods, `diameter_method = "contour"` (default) or `"edt"`, measured on a local normal (`_local_normal`, `_measurement_vector`, `diameter_vector_half_length_m = 1.0`) — so “perpendicular to the centerline” is confirmed.

Volume: `calc_l_v(p1, p2, d1, d2)`, truncated cone between consecutive nodes. Confirmed.

3.6 6. Node spacing is 50 cm, not 25 cm

The WINMOL documentation says the diameter is determined “every 25 cm”. The code says:

```
measuring_point_spacing_m = 0.5 # Config.py:104
measuring_point_spacing = math.floor(
    min(config.min_length, config.measuring_point_spacing_m) / px_size) # Skeletonization.py
```

`min(2.0, 0.5) = 0.5` -> **50 cm**. Measured on real output across four orthomosaics, the mean gap between consecutive centreline vertices is **0.47–0.50 m**. Either the docs describe a different version or the figure is wrong; the shipped default is 50 cm.

3.7 7. Two coordinate traps in the vectoriser

Both produce plausible geometry rather than an error, so they are invisible without checking:

- **The skeletonisation pad is never removed.** `find_segments` pads by `max_tree_height / px_size` and returns coordinates still carrying that offset — 1530 px at 2.09 cm/px. The comment at `Skeletonization.py:175` claims the offset is removed; `np.argmax` returns raw padded indices.
- **Paths are pixel (row, col), not world coordinates,** and `path.length` is a pixel count. World coordinates are applied only at write time. `calc_v_d_edt` passes `stem.path.coords` to `_xy_to_rowcol`, which reads them as easting/northing — so the **EDT diameter branch appears broken**; the default contour path is self-consistent in pixel space.

3.8 8. CLI: Stems does not vectorise

`run_stem_pipeline` calls the prediction phase and stops; only `run_tree_pipeline` (Trees) runs `run_vector_phase`. A Stems run exits cleanly with a stem raster and no vectors — which reads as a failure and is not one.

3.9 What this changes in the report's recommendations

Stage 1 (data). Target the code's actual scale — 15 m / 512 px = 2.93 cm/px — rather than “3 cm”. Our measurement: serving a model at a GSD it was not trained on cost 14.6 F1, and the same published beech model scores **0.8003** on native-512 tiles versus **0.6812** on Reder's own 313 px TestDS (upsampled $\times 1.64$ to reach the input). Resolution handling, not model quality, explains a 12-point swing.

Stage 2 (splits). The report's advice to use site-level holdout is right and the code does not do it. Worth knowing what it costs: in our corpus a whole-site holdout costs 8–11 F1, and one site (November beech canopy, 100% amber pixels) returns F1 **exactly 0.0000** — so a leave-one-site-out mean can be carried entirely by one dead fold. Report per-fold.

Stage 2 (copy-paste + two-stage). GenDS tiles are 2000 $\times$ 2000 but carry ~500–650 px of real information — round-trip loss through 512 is 1.45 grey levels at 2 $\times$ against SpecDS's 3.73, and edge energy is 7.97/px against 15.52. Effective GSD $\approx$ 4.6 cm/px, which matches SpecDS's 4.8 cm/px, so the two stages are well aligned with each other but not with the Analyzer default. On our own corpus, synthetic pretraining measured **-0.2 to -1.4** once a validation-split bug was fixed (it had read +1.1 to +3.6 before).

Stage 4 (evaluation). Pixel F1 is not comparable to DR25 — different quantity. Also fix a noise floor before comparing anything: two runs with **bit-identical gradients** landed 2.0 F1 apart without deterministic kernels, 0.5–1.0 with. Single-run differences below that are unreadable, which is why the ablations above are paired over 5 seeds.

3.10 A reproduction worth noting

The 2022 paper reports pixel F1 **72.6%** for SpecDS-only training and **75.6%** for its best two-stage pretrained model, on TestDS.

Trained single-stage on the same published SpecDS, scored on the same published TestDS with our own metric code: **0.7581**. That matches the paper's *best pretrained* result without any GenDS pretraining. The R U-Net retrained under the same conditions gives 0.7424.

Two caveats. The published **beech** model (SpecDS_Beech_512, 2023-02-28) scores only **0.6812** on that same TestDS — below the 2022 paper's figure, unexplained, and worth raising with the author. And TestDS as distributed on Zenodo contains **117** tiles, not the 106 stems the paper describes (verified by md5 against the record).

Sources read directly: `WINMOL_segmentor` — `controlling.R`, `training.R`, `callbacks.R`, `main_training.R`, `model_UNet.R`, `input_pipeline.R`. `WINMOL_Analyzer` — `Config.py`, `Skeletonization.py`, `Vec-`

torization.py, *Quantification.py*, *winmol_run.py*. Datasets verified byte-identical against Zenodo record 5682580.

4 Scale augmentation: $\pm 30\%$ footprint jitter buys robustness for free

Design and yardstick were fixed before training — [superpowers/specs/2026-08-13-scale-augmentation-design.md](#). Numbers below are copied verbatim from [assets/scale-sweep.json](#), rendered by `scripts/plot_scale_sweep.py` from the six per-model sweeps in `assets/scale-aug/`.

Answer: ship it. Measured on two architectures. On HRNet, jitter flattens the scale curve by **0.87 F1** (3/3 seeds, paired $t = -6.02$) and costs **0.11 F1** at the native scale — 1/3 seeds positive, i.e. noise. On UNet the same manipulation flattens it by **1.88 F1** (3/3 seeds, $t = -56.95$) and *gains* 0.20 F1 at native scale as well. It wins at every scale on every UNet seed.

Sections below cover HRNet first, then the UNet replication and a reproducibility check.

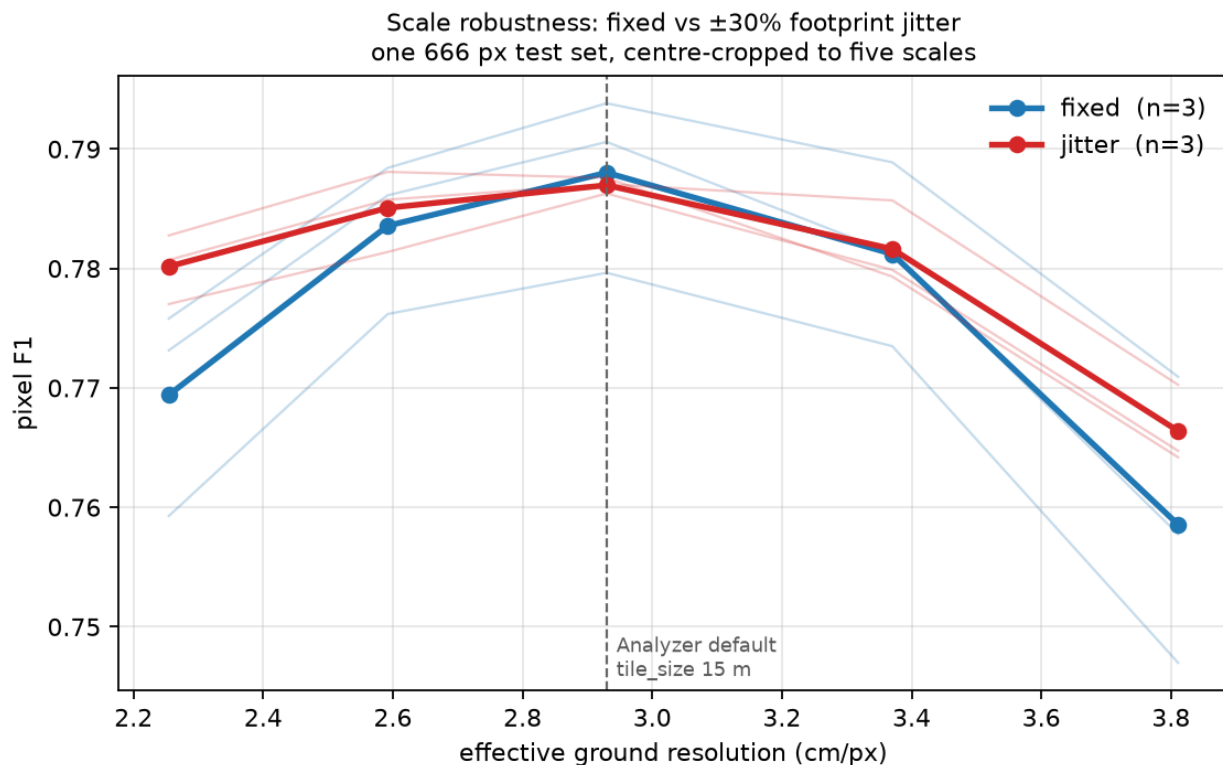


Figure 1: F1 versus effective ground resolution

4.1 Setup

data	BeechScale666 — 3388 / 400 / 400 tiles at 666 px, 19.512 m, 2.9297 cm/px
sites	Campus + Oberheide block-split (60 m); Bachsee_north, Kaufland whole-to-train
leak check	closest train<->val<->test approach 28.3 m / 27.9 m vs a 27.59 m floor

arms	crop 512–512 (<i>fixed</i>) vs 394–666 (<i>jitter</i>), RandomSizedCrop->512
held fixed	same tiles, rotation/flip/photometric, arch, batch 16, --deterministic
n	3 paired seeds per arch (HRNet, UNet); test = 400 tiles, CPU EP, threshold 0.5

Both arms train on **the same 666 px source tiles** and both get random crop *position* — only the scale distribution differs. That is what makes the comparison clean.

4.2 HRNet: per-scale result

Differences are jitter – fixed, paired within seed.

crop	ratio	GSD	fixed	jitter	diff	sign	t
394	0.77×	2.254	0.7694	0.7802	+0.0108	3/3	2.87
453	0.88×	2.592	0.7836	0.7851	+0.0015	2/3	0.66
512	1.00×	2.930	0.7880	0.7870	–0.0011	1/3	–0.26
589	1.15×	3.370	0.7812	0.7816	+0.0004	1/3	0.14
666	1.30×	3.811	0.7586	0.7664	+0.0078	2/3	1.46

Drop from each arm’s own peak — the robustness claim, independent of absolute level:

arm	s1	s2	s3	mean
fixed	0.0229	0.0326	0.0328	0.0295
jitter	0.0168	0.0215	0.0239	0.0207

Paired: **–0.0087, 3/3 seeds flatter, t –6.02.**

4.3 Reading it

The win is at the flanks and it is a recall win. At 1.30× the fixed arm’s precision *rises* to 0.819–0.829 while recall falls to 0.680–0.728: coarser pixels make stems thinner, the model stops committing, and it under-segments. This is the benign failure mode — an under-reporting model, invisible in F1 alone. Jitter holds recall 3–6 points higher there at some precision cost. Every domain-shift failure measured in this project has had this same shape.

The per-scale t-values are weak; the spread test is not. With n=3, only the 0.77× point approaches significance on its own. But the spread statistic uses each seed’s whole curve rather than one point of it, and it is unanimous with |t| = 6.0. The per-scale rows are best read as *where* the effect lives, not as five independent tests.

The cost at native scale is nil. –0.0011 F1, 1/3 seeds positive. The worry going in — that spreading training over 2.25–3.81 cm/px would blunt the model at the one resolution it is actually served at — did not materialise. Note also that validation ran at 1.00× (--eval-tiling), which structurally favours the fixed arm, so this is if anything generous to the baseline.

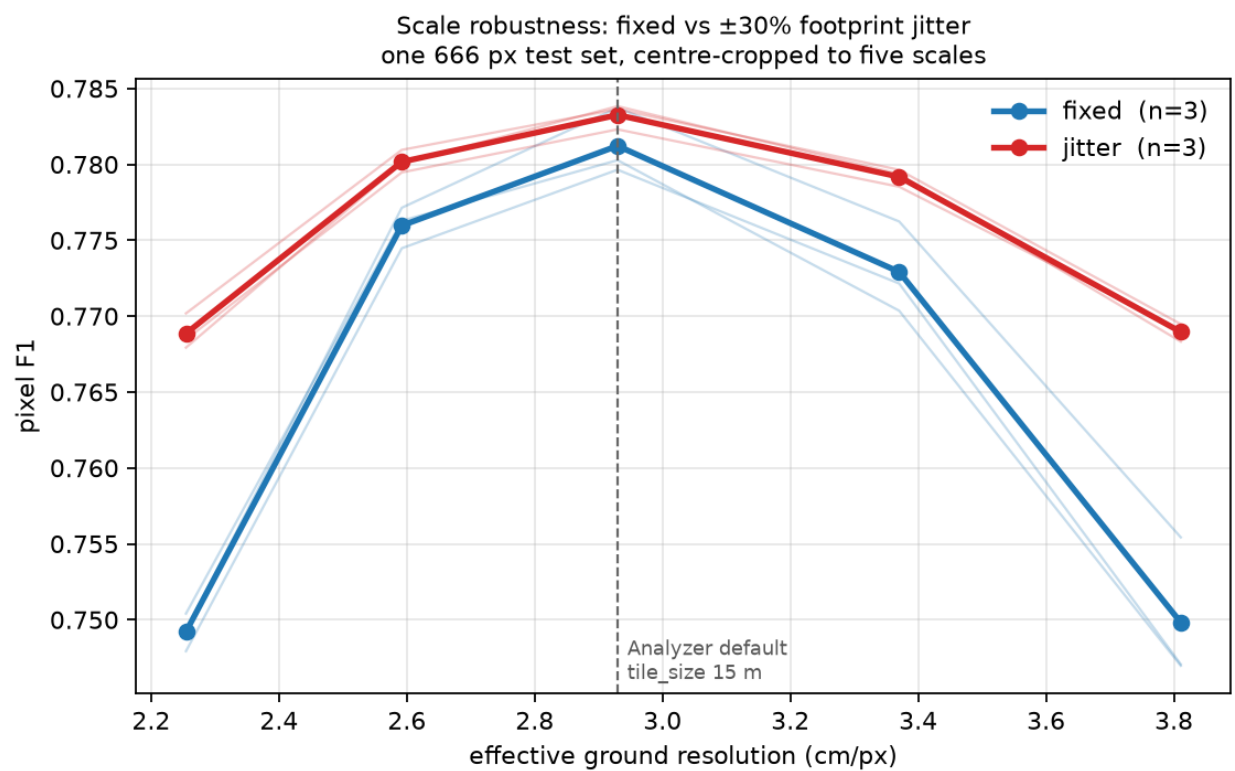


Figure 2: UNet: F1 versus effective ground resolution

4.4 Replication on UNet — the effect is real and larger

Repeated identically with `--arch unet` (same dataset, crop ranges, seeds, everything); sweeps in [assets/scale-aug-unet/](#), aggregate in [assets/scale-sweep-unet.json](#).

crop	ratio	GSD	fixed	jitter	diff	sign	t
394	0.77×	2.254	0.7492	0.7689	+0.0197	3/3	14.97
453	0.88×	2.592	0.7760	0.7802	+0.0042	3/3	3.67
512	1.00×	2.930	0.7812	0.7833	+0.0020	3/3	1.82
589	1.15×	3.370	0.7729	0.7792	+0.0063	3/3	3.97
666	1.30×	3.811	0.7498	0.7690	+0.0191	3/3	6.13

Spread: fixed **0.0334** (0.0344/0.0333/0.0326) vs jitter **0.0146** (0.0159/0.0138/0.0142). Paired: **−0.0188, 3/3 seeds flatter, t −56.95**.

Jitter wins at **every** scale on **every** seed, and unlike HRNet it does not even cost anything at 1.00× — it gains 0.20 F1 there, 3/3 seeds.

Why UNet gains more. HRNet carries parallel branches at several resolutions through the whole network and fuses them repeatedly, so multi-scale context is architectural. A plain UNet has one resolution ladder and must *learn* scale invariance from the data. That predicts exactly what is measured: UNet is the more scale-brittle baseline (spread 0.0334 vs HRNet’s 0.0295) and gains the most from jitter (−0.0188 vs −0.0087). The two results agree in direction and differ in magnitude in the direction the architectures predict, which is a stronger joint result than either alone.

	fixed spread	jitter spread	gain	cost at 1.00×
HRNet	0.0295	0.0207	−0.0087 (t −6.0)	−0.0011 (1/3, noise)
UNet	0.0334	0.0146	−0.0188 (t −57.0)	+0.0020 (3/3)

4.5 Reproducibility check

fixed-s1 scored highest in its arm in both architectures, so it was re-run from scratch with the same seed on a different GPU:

	F1	precision	recall
original (GPU 0)	0.7922	0.7990	0.7856
reproduction (GPU 7)	0.7922	0.7990	0.7856

Those are `test_results.md` figures — 1600 windows from `--eval-tiling`, not the sweep’s 400 centre crops, so they differ slightly from the 1.00× column above (0.7922 vs 0.7938 for the same model). Every arm-vs-arm number in this report comes from the sweep; the table here is a run-identity check, not a scale measurement.

Identical to four decimals. `--deterministic` reproduces across devices, so between-seed differences of 1–2 F1 in the fixed arm are genuine seed variance, not run-to-run noise — and being common to both arms, they cancel in the paired differences. Runs now write `run_config.json` beside the model so this is auditable from the artifacts rather than requiring a re-run.

4.6 Caveats, stated

- **F1 is not comparable across scales.** A 23 cm stem is 10.2 px at 2.25 cm/px and 6.0 px at 3.81. The downward slope on the right of the figure is partly the task getting harder, not the models getting worse. Only the arm-vs-arm gap within a column is readable.

- **Arm B occasionally sees the full 666 px tile; arm A never does.** A wider crop range means a wider field of view. This is inherent to the manipulation and cannot be separated from scale jitter without also changing the footprint.
- **This is not the earlier “5.2 F1 across $\pm 30\%$ zoom” figure.** That came from a different model and a test built by re-sampling the orthomosaic per scale, which put each scale on different ground. The 2.95 F1 (HRNet) and 3.34 F1 (UNet) measured here are the clean version of the same quantity and supersede it; the two are not directly comparable.
- **Sites are shared across splits.** The claim is scale robustness within known forests, not transfer to a new site.
- **n = 3.** Enough for the unanimous spread result, thin for the per-scale rows.

4.7 Recommendation

Enable `--multiscale --crop-min-px 394 --crop-max-px 666` by default for beech training, on both architectures. It removes **30%** of the scale degradation on HRNet and **56%** on UNet, and costs nothing at `tile_size = 15` — on UNet it gains there too. Users do move the tile-size spinbox, and nothing in the model warns them when they have.

This does **not** replace matching the training scale to the serving scale. That was worth 14.6 F1; this is worth 0.9–1.9. Be robust *and* correctly scaled.

Two questions worth answering next, in order:

1. **Does a wider range keep paying?** The corpus spans 1.58–6.39 cm/px, a 4 $\times$ range, against the 1.7 $\times$ tested here. The UNet curve is still falling at both ends of the sweep, so $\pm 30\%$ is unlikely to be where the gain stops.
2. **Does it survive a site holdout?** These splits share sites. **Answered** — see [scale-augmentation-los0.md](#). Flatter in 4/4 folds, and worth **+3.3 F1 at the deployment scale** on the one clean holdout where the model segments at all. Two of the four folds turned out not to be site holdouts (Campus and Campus_Oberheide are the same forest 4.4 years apart, 33% footprint overlap) and one was dead (Bachsee_north, F1 < 0.13 for every arm), so the new-site evidence is one strong fold rather than four.

5 Scale jitter under a site holdout — four folds, reported individually

Follow-up to [scale-augmentation-results.md](#), which measured scale jitter on splits that **share sites**. This asks whether the robustness survives when the test site is absent from training. UNet, 3 paired seeds per arm per fold, 24 runs. Numbers copied verbatim from `assets/los0-*.json`.

Answer: it survives, but the evidence is thinner than the within-site result and one fold is uninformative. Jitter’s scale curve is flatter in **4 of 4 folds**; only one fold reaches the usual significance bar on its own.

5.1 Two of these folds are not site holdouts

`tiles.jsonl` says **33% of Campus tiles overlap a Campus_Oberheide footprint** (centroids 183 m apart). They are the same forest imaged 2017-10-16 and 2022-02-26, and windthrown stems do not move between acquisitions — so those are largely the same physical stems.

fold	test tiles	nearest training site	what it measures
Bachsee_north	800	5.68 km	new-site transfer
Kaufland	388	3.56 km	new-site transfer
Campus	800	0 m (33% overlap)	4.4-year acquisition gap
Campus_Oberheide	800	0 m (30% overlap)	4.4-year acquisition gap

Reported under those two headings, never pooled.

5.2 Every fold, no mean

spread = each seed's peak-minus-worst F1 across 0.77× / 1.00× / 1.30×. Lower is more scale-robust; the paired column counts seeds where jitter is flatter.

fold	F1@1.00× fixed	jitter	spread fixed	spread jitter	flatter	t
Bachsee_north	0.0945	0.0434	0.0940	0.0778	1/3	−0.15
Kaufland	0.6737	0.7065	0.1231	0.1116	2/3	−1.31
Campus	0.5263	0.5290	0.1125	0.0886	2/3	−1.41
Campus_Oberheide	0.6793	0.6792	0.0408	0.0272	3/3	−5.52

Direction is unanimous — all four mean spreads favour jitter — but only Campus_Oberheide clears |t| ≥ 3 with unanimous seeds. State this as a fold count, not a mean: *flatter in 4/4 folds, individually significant in 1/4.*

5.2.1 Bachsee_north is a domain-shift fold, not a defective one

F1 **0.03–0.13** for every arm and seed. Three explanations were tried; the first two are wrong and are recorded here because each looked convincing.

Not a labelling error. Its annotations are EPSG:25833 and its orthomosaic EPSG:32633 — the same UTM zone on different datums, a pair that has diverged ~0.5 m through plate motion, and already logged as a known defect. Measured with `scripts/check_registration.py` on the orthomosaic's **own unrotated grid**, every beech site peaks at zero offset:

site	stems / ortho CRS	contrast at zero	bright peak	verdict
Bachsee_north	25833 / 32633 mismatch	+0.118	(0.00, 0.00) m	registered
Campus	32633 / 32633	+0.384	(0.00, 0.00) m	registered
Campus_Oberheide	25833 / 25833	+0.811	(0.00, 0.00) m	registered
Kaufland	25833 / 25833	+0.797	(0.00, 0.02) m	registered

The pipeline's `t transform_geom` applies the datum shift; residual under 4 cm.

Not invisible stems either — this was my error. The table above measures *luminance*, and on that basis Bachsee looked like a closed canopy hiding its stems. It is not. Its stems differ from the background in **colour**, not brightness. In CIELAB over 100 random tiles per site:

site	luminance sep	chroma sep	full Lab sep	fold F1
Kaufland	0.77	1.98	2.14	0.71
Bachsee_north	0.44	1.66	1.75	0.09
Campus_Oberheide	0.56	1.51	1.70	0.68
Campus	0.53	0.95	1.16	0.53

Bachsee has the **second-strongest** stem/background separation in the corpus. Campus has the weakest and scores 0.53. An earlier claim in this repo that fold F1 is monotone in stem contrast came from the luminance-only statistic and is **withdrawn**.

Colour domain by site — random sample of 4, seed 1

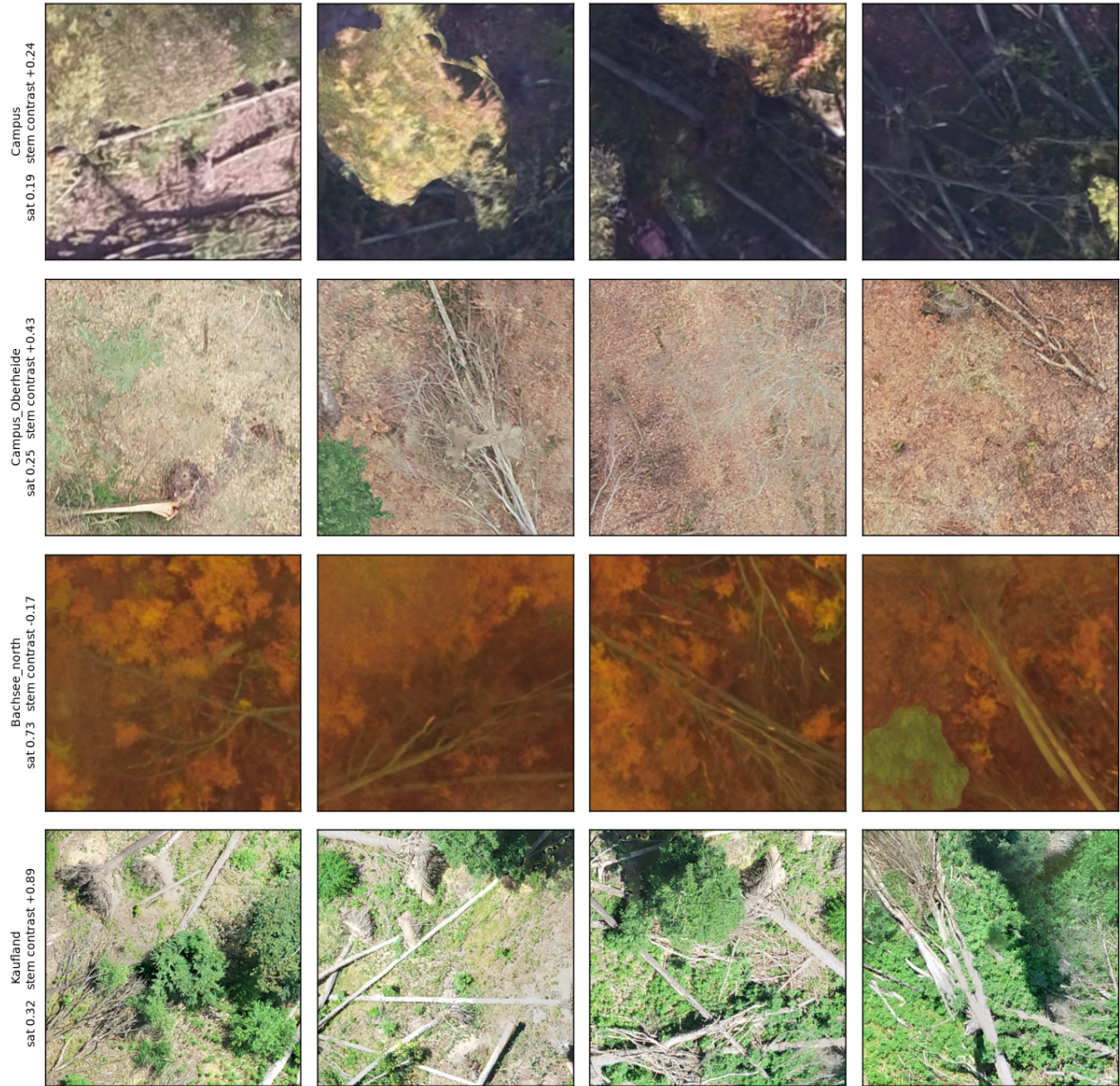


Figure 3: Colour domain by site



Figure 4: Bachsee stem close-ups

The close-ups make it plain — pale olive stems against orange canopy, labels sitting on them:

It is domain shift, and one training run proves it. Trained on 252 tiles from the west half of Bachsee and tested on the spatially separated east half (30.3 m closest approach against the 27.59 m floor):

training data	test	F1	precision	recall
3,588 tiles, other sites (the LOSO fold)	Bachsee	0.09	—	—
252 tiles, Bachsee's own west half	Bachsee east	0.6203	0.6061	0.6351

252 of its own tiles beat 3,588 of everyone else's by 53 F1 points. The signal is there, the labels are on it, and the site is learnable. It fails as a holdout because it is the corpus's **only autumn-phenology site**: held out, nothing in training has ever shown the model a stem against orange foliage.

Green is ground truth, red is prediction; on most tiles there is no red at all — recall collapse in its extreme form. The fold remains a **tie** for arm comparison, but for a different reason than “hard data”: both arms are equally outside their training domain.

5.2.1.1 Strong hue augmentation fixes it — +64.6 F1 If Bachsee fails only because no training site is amber, randomising hue should remove the colour shortcut and force the model onto structure. Tested: 12 UNet runs, 3 paired seeds, arms differing **only** in HueSaturationValue limits, both re-run so nothing is inherited.

arm	hue	sat	val	p
base	±20	±30	±20	0.5
strong	±90 (any hue)	±60	±30	0.9

fold	base	strong	delta	signs
Bachsee_north (out of domain)	0.042	0.688	+0.646	3/3
Kaufland (in domain)	0.708	0.690	−0.017	0/3

Per seed on Bachsee: 0.0000 / 0.0639 / 0.0619 -> 0.6660 / 0.7059 / 0.6916.

The failure mode inverts. The base arm's precision is 0.87–1.00 with **recall 0.00–0.03** — it predicts essentially nothing. Strong moves recall to 0.58–0.66 at precision 0.76–0.79. Note 0.688 also exceeds the 0.62 that training on Bachsee's *own* west half achieved, so the other three sites do contain the information; the model simply could not reach it through the colour gap.

Cost: −1.7 F1 in domain (3/3 seeds, |t| 1.6). That is a real cost and small enough to be worth paying for anything expected to meet a new acquisition or season.

5.2.1.2 It is an interaction, not a dose — milder settings fail base and strong differ in four knobs at once, so two further arms decompose it (24 runs total, same 3 seeds):

Bachsee_north (held out, F1 < 0.13) — random sample of 5, seed 1

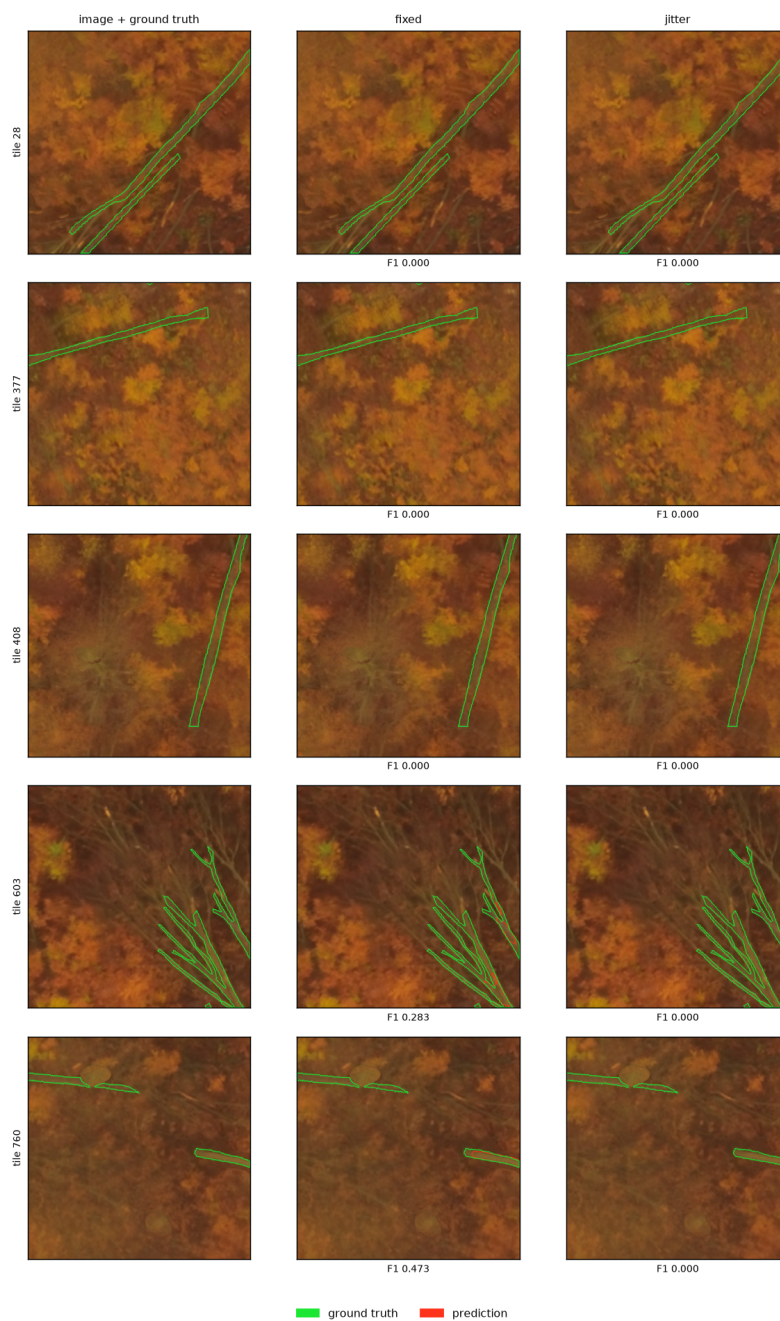


Figure 5: Bachsee_north examples

arm	hue	sat	val	p	Bachsee	delta vs base	Kaufland	delta
base	±20	±30	±20	0.5	0.042	—	0.708	—
mid45	±45	±60	±30	0.9	0.177	+0.135	0.705	−0.002
hueonly	±90	±30	±20	0.5	0.261	+0.219	0.707	−0.001
strong	±90	±60	±30	0.9	0.688	+0.646	0.690	−0.017

Neither ingredient works alone. A **full hue circle at the default probability** reaches only 0.261, and **half the hue range with the full probability and saturation** reaches 0.177 — while both together reach 0.688. The gain is roughly 3× what either part contributes, so this is an interaction and not a dose-response curve: the model needs both a hue range wide enough to reach amber *and* a probability high enough that it cannot fall back on the unshifted half of the data.

Per-seed Bachsee: mid45 0.203/0.113/0.216, hueonly 0.182/0.428/0.172 (note the spread — 0.428 on one seed, so hueonly is unstable as well as insufficient).

The in-domain cost scales with the same combination: −0.001 (hueonly), −0.002 (mid45), −0.017 (strong). Buying the +0.646 costs 1.7 F1; the cheaper settings save nothing worth having because they do not deliver the gain.

Recommendation: `--aug-hue-shift 90 --aug-sat-shift 60 --aug-val-shift 30 --aug-hsv-p 0.9`. Do not soften it.

This retires the standing advice that a site holdout is impossible on this corpus. Untested: whether this transfers to spruce and pine, and whether p alone (hue ±20 at p 0.9) does anything — the two arms here vary p together with sat. Failing that, this site should never be the held-out one.

5.2.2 Kaufland is the informative clean holdout

The only true spatial holdout where the model works at all, and jitter wins **at every scale on every seed**:

ratio	GSD	fixed	jitter	diff	sign	t
0.77×	2.254	0.5786	0.6237	+0.0451	3/3	4.37
1.00×	2.930	0.6737	0.7065	+0.0328	3/3	2.61
1.30×	3.811	0.7017	0.7353	+0.0336	3/3	9.58

Here jitter is worth **+3.3 F1 at the deployment scale** — an order of magnitude more than the +0.2 measured within-site. Note the curve *rises* with coarseness on this fold, unlike every other: Kaufland is the corpus’s finest orthomosaic (2.09 cm/px native) and its densest (stem fraction 0.063 against 0.028–0.038), so its peak sits outside the sampled range. Its spread is therefore a lower bound.

5.3 The curves

Per fold, F1 against effective ground resolution, per-seed spread drawn rather than summarised. Note the y-scales differ by an order of magnitude between folds.

Kaufland (held out, the informative fold) — random sample of 4, seed 1

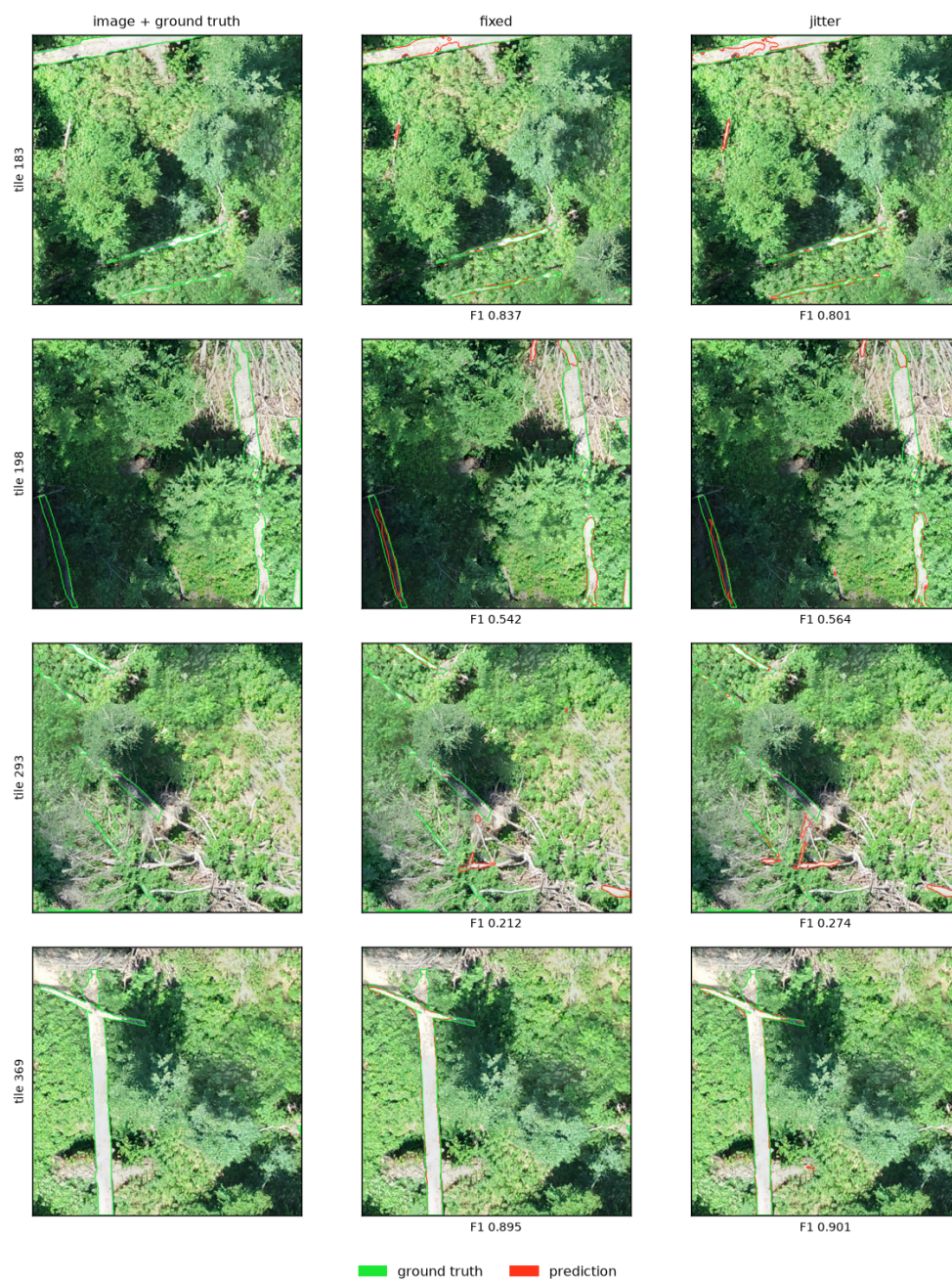
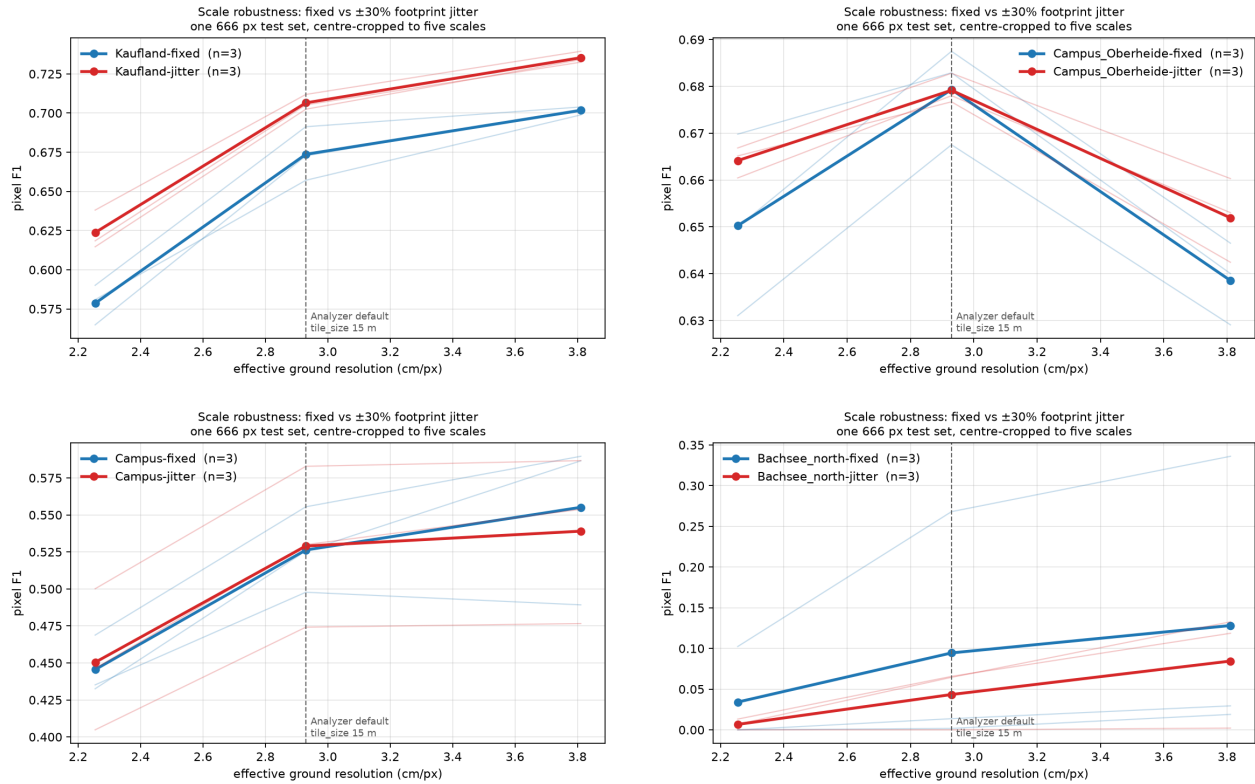


Figure 6: Kaufland examples



5.4 Does Bachsee_north belong in training? Yes — tested

Before it was understood as domain shift, Bachsee looked like 800 tiles of unlearnable noise sitting in ~20–29% of every training set. So the folds were re-run with Bachsee **excluded from training entirely** (3 folds x 2 arms x 3 seeds = 18 UNet runs, everything else identical). Paired within fold, arm and seed, at 1.00x:

fold (acquisition)	train tiles	F1 with Bachsee	without	delta	signs
Kaufland (Jul, summer green)	4000 -> 3200	0.6901	0.6953	+0.005	5/6
Campus_Oberheide (Feb, leaf-off)	2788 -> 1988	0.6793	0.6338	-0.045	0/6
Campus (Oct, autumn)	2788 -> 1988	0.5277	0.4324	-0.095	0/6

Overall **-0.045 F1**, 5 of 18 paired comparisons positive. **Removing Bachsee hurts.**

Declared confound: dropping the site also drops 800 training tiles, so Campus and Campus_Oberheide lose 29% of their data and Kaufland 20%. Quantity alone would predict the direction. It does not predict the *ordering*: Campus and Campus_Oberheide lose the **identical** 800 tiles from identical 2788-tile sets, yet Campus loses twice as much F1. What separates them is phenological distance from Bachsee's November

amber — Campus was flown in October, Campus_Oberheide in leaf-off February, Kaufland in high summer, and the damage falls in exactly that order.

So the out-of-domain site is not noise to be removed; it is the nearest thing the corpus has to phenological diversity, and the sites that resemble it depend on it most. In a corpus with under 1 ha of labelled stem, discarding an acquisition is expensive.

The arm comparison is unaffected in direction: jitter’s curve stays flatter in 3/3 folds without Bachsee (-0.0142, -0.0450, -0.0012), though at 1.00x it now loses on the two Campus folds and wins only on Kaufland — consistent with those folds simply having less data to work with.

5.5 What the holdout costs

	within-site (pooled blocks)	site holdout
F1 @ 1.00x	0.78	0.04 – 0.71

Between 7 and 74 F1 points, depending entirely on which site. Any single-fold number here would have been arbitrary — which is the argument for reporting all four.

5.6 Caveats

- **n = 3 seeds per arm per fold.** Under domain shift the within-arm seed spread reaches **12 F1 points** (Campus fold), against 1–2 within-site. Absolute F1 differences below ~3 points are unreadable at this n; the spread statistic is within-model and survives better, which is why the claim rests on it.
- **Only 2 of 4 folds are true spatial holdouts**, and one of those is dead. The new-site evidence is effectively **one fold** — strong, but one.
- **Train sizes differ by fold** (2788–4000 tiles), inherent to leave-one-site-out.
- **Val comes from one site** in the Campus and Campus_Oberheide folds, since only those two sites are large enough to block-split at a 19.512 m footprint.
- **Corpus limit.** Four beech sites, two of them the same ground. A stronger claim needs annotated sites this corpus does not have — see winmol-label-scarcity.

5.7 Verdict

The within-site recommendation stands and is mildly strengthened: jitter never hurt in any fold, flattened the curve in all four, and on the one clean holdout where segmentation works at all it was worth +3.3 F1 at the deployment scale. But calling this “validated across sites” would overstate a corpus with one informative holdout fold.

6 Working process

6.1 Splits

Standard: no tile footprint may overlap between train, val and test. Spatial autocorrelation inside a split is accepted and expected.

`make_splits.py --strategy blocks` satisfies this by construction: the AOI is cut into a grid, whole blocks are dealt to splits, and each block is then shrunk by the footprint half-diagonal. Two centres on opposite sides of a shared edge are therefore at least extent $\cdot \sqrt{2}$ apart — the distance below which two rotated footprints can share a pixel.

Verification is a pass/fail check, not a discussion. `scripts/plot_splits.py` prints the measured closest centre-to-centre distance per site and exits non-zero if a site is missing from a split. Run it after every build; report the number once.

	requirement
footprint overlap between splits	none — closest centre distance > extent · $\sqrt{2}$
spatial autocorrelation within a split	accepted
every site in every split	yes, unless the AOI is too small to cut (record in <code>splits.json</code>)
provenance	<code>tiles.jsonl</code> per split: site, centre, angle, GSD, stem fraction

A site whose AOI cannot yield 12+ usable blocks goes wholesale into one split via `whole_split`, recorded in `splits.json`. No further comment needed.

6.2 Scale

Train at the scale the Analyzer serves. Effective GSD is `extent_m / tile_px` in training and `tile_size / 512` in the Analyzer. These must match, or the model is served at a different resolution than it learned.

Default `tile_size = 15` -> **2.93 cm/px** -> build with `--extent 15 --tile-px 512`.

Measured on Kaufland, AOI-masked, same model family:

trained at	served at	F1
2.00 cm/px	2.93 cm/px (mismatched)	0.7389
2.00 cm/px	2.00 cm/px (matched)	0.8091
2.93 cm/px	2.93 cm/px (matched, default)	0.8844

Ship the `tile_size` alongside any model that is not trained at 2.93 cm/px.

Jitter the training footprint $\pm 30\%$ — `--multiscale --crop-min-px 394 --crop-max-px 666` on a 666 px source. Measured on two architectures, 3 paired seeds each: it flattens the F1-vs-GSD curve by 0.87 F1 on HRNet and 1.88 on UNet, and costs nothing at the native scale. See [scale-augmentation-results.md](#). Matching the scale is still worth far more than being robust to it — do both.

6.3 Evaluation

- Tile-level F1 for comparing training runs.
- **Full-orthomosaic inference masked to the AOI** for anything claiming deployment performance. Outside the windthrow polygon stems are real but undigitised; scoring the whole raster is meaningless (measured: 0.2974 vs 0.8844 on the same prediction).
- Report precision and recall separately.

6.4 Reporting

- Numbers come from each run's own `test_results.md`, collected by `scripts/collect_results.py` into JSON. Reports render from that JSON.
- State the test set and the training scale beside every figure.
- One line per caveat, in a caveats section. Not inline.

6.5 Pipeline order

`fix` -> `sample` -> `split`, enforced by `make_splits.py`. Geometry repair happens first because GEOS aborts on the first set operation touching an invalid ring.

6.6 Environments

what	where
corpus	/Volumes/storage/Datasets/Winmol/training_data/WINMOL
derived tiles	data/ in this repo, git-ignored, regenerable from configs/*.json
training	carrot /raid/cwinkelmann/winmol — check GPUs are free before launching
Analyzer	~/hnee/WINMOL_Analyzer, needs Python ≥ 3.10